John Vhen Ocsona
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EDUCATION

California State Polytechnic University, Pomona ***August 2025 – May 2029***
Bachelor of Science, Mechanical Engineering (GPA: 3.57)
Related Coursework: *MATLAB, SolidWorks, Vector Statics, Physics I*

SKILLS

Software: SolidWorks, AutoCAD, Microsoft Office, Google Workspace, MATLAB, Python, C++, Arduino, Ansys
Fabrication: Welding (SMAW, GMAW, FCAW), 3D Printing, CNC Cutting, Wood and Metal Machining, Circuit Prototyping, Foam Cutting

RESEARCH & PROJECT EXPERIENCE

Autonomous Vehicle Laboratory (AVL), California State Polytechnic University, Pomona
All-Terrain Autonomous Vehicle Research Project ***January 2026 – Present***

- Modeled a 2-DOF suspension system in MATLAB, modeling body acceleration, tire deflection, and suspension travel, to compare passive, active, and semi-active control strategies.
- Designed a double wishbone suspension assembly in SolidWorks, including fit verification and design validation.
- Developed and designed a new battery mount, improving maximum force endurance and stability.
- Reduced RMS body acceleration by 15.1% on step-bump input and 5.2% on washboard input using active control versus passive suspension.

Project Pegasus (Lockeed Martin-sponsored), California State Polytechnic University, Pomona
55 lb & 110 lb Electronic Vertical Take-Off and Landing Vehicle Project ***March 2026 – Present***

- Performed system test and evaluation documentation for a heavy-payload eVTOL propulsion thrust-stand test while following test safety protocols.
- Performed failure analysis on a crash, analyzing damage across 30 components to support post-test evaluation and follow-up actions.
- Designed and 3D printed landing gear and structural mounts, reducing weight and improving shock absorption during high-payload testing.
- Designed and 3D printed a tail cone in SolidWorks for an eVTOL vehicle design, optimizing its aerodynamic shape and capabilities.
- Manufactured a lightweight vehicle fuselage using foam shaping and 3D printed components.

First-Year Experience PolyX, California State Polytechnic University, Pomona
Team Lead, Automatic License Plate Recognition (ALPR) Research Project ***February 2026 – Present***

- Led a team of students in researching parking permit enforcement issues at CPP and proposing an ALPR-based solution concept.
- Researched comparable ALPR implementations to assess potential impacts on enforcement efficiency, parking access, and permit revenue.
- Presented the team's solution at a campus fair, explaining the problem, proposed system, and implementation tradeoffs.
- Coordinated project planning, task delegation, progress checks, meeting scheduling, and milestone tracking.